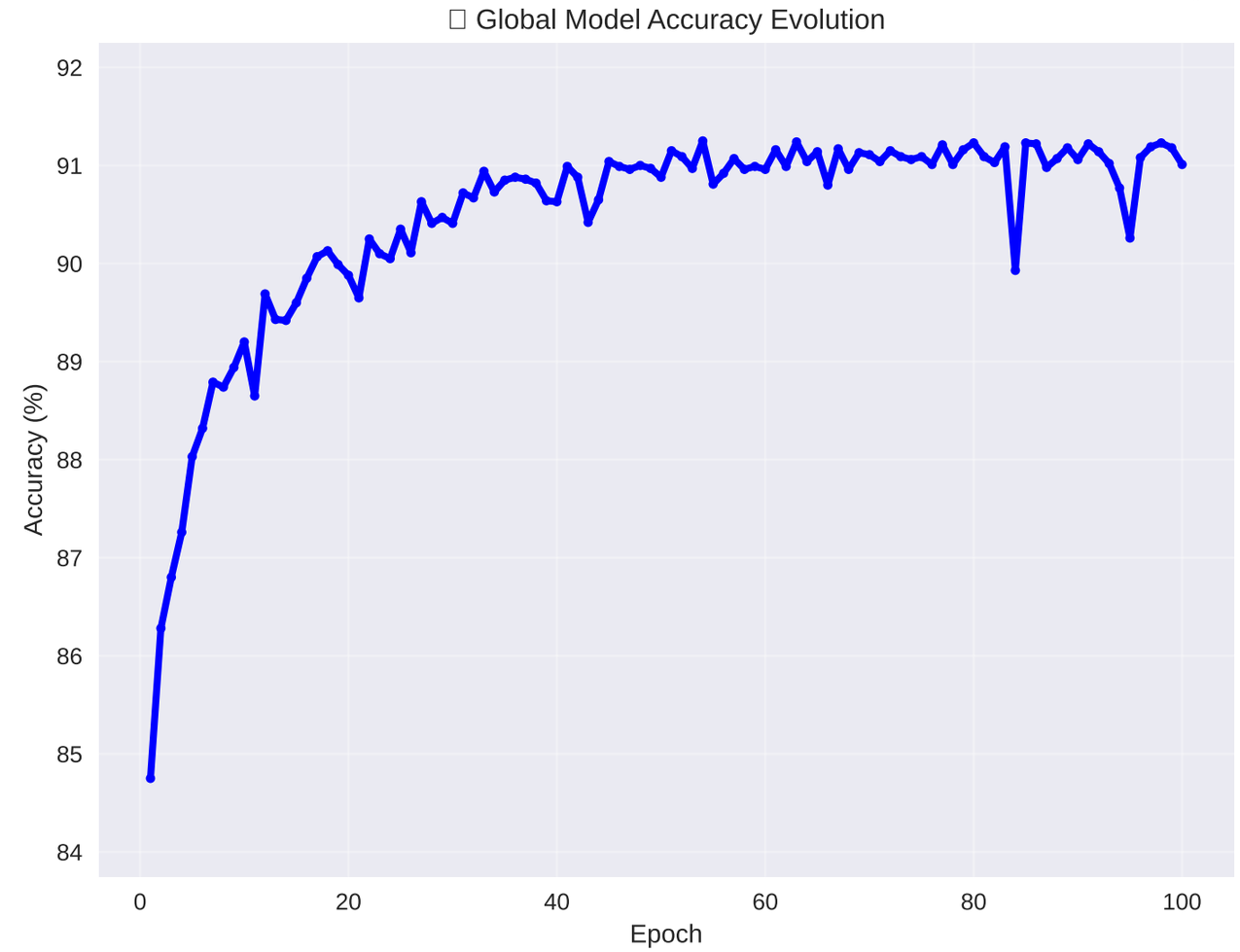
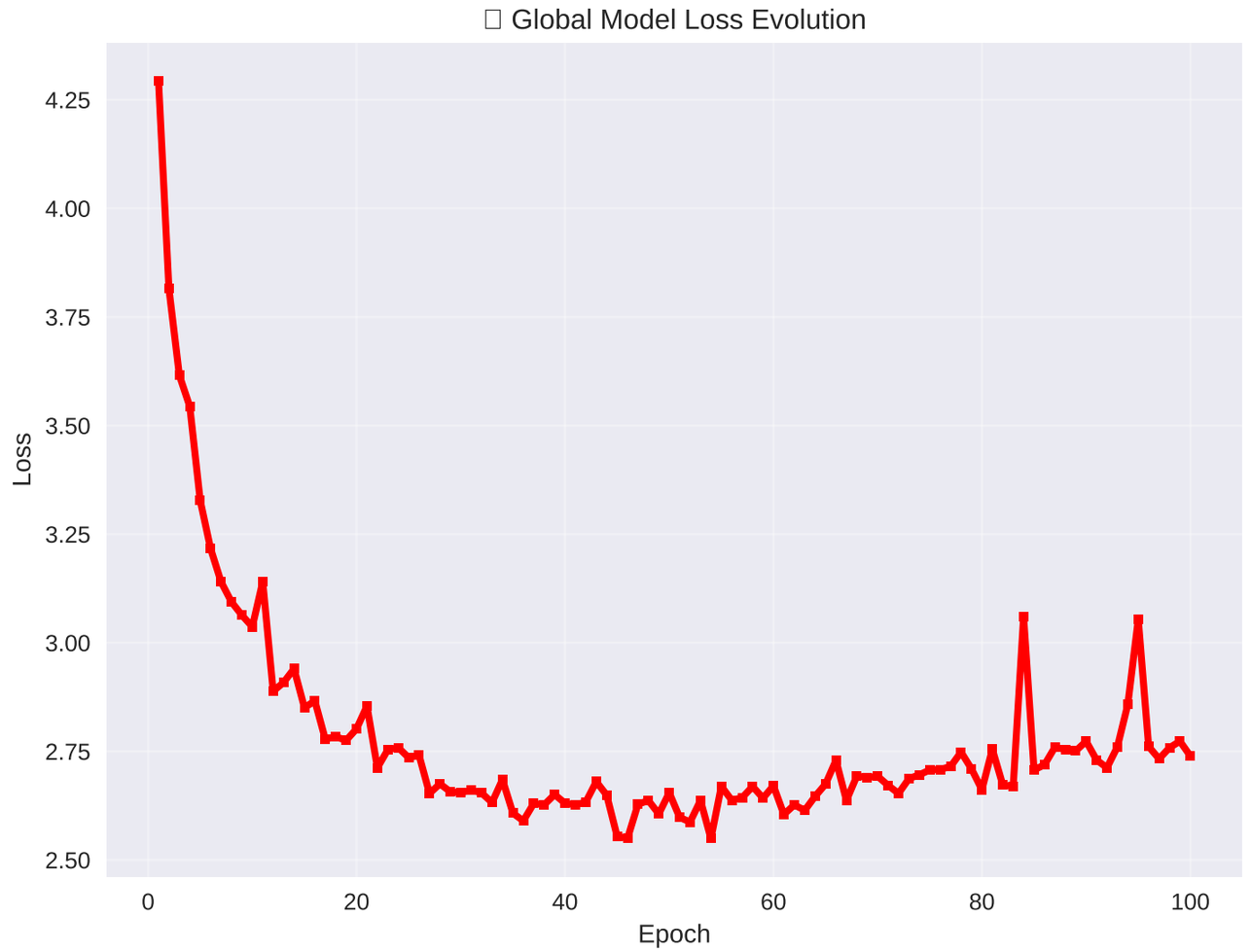
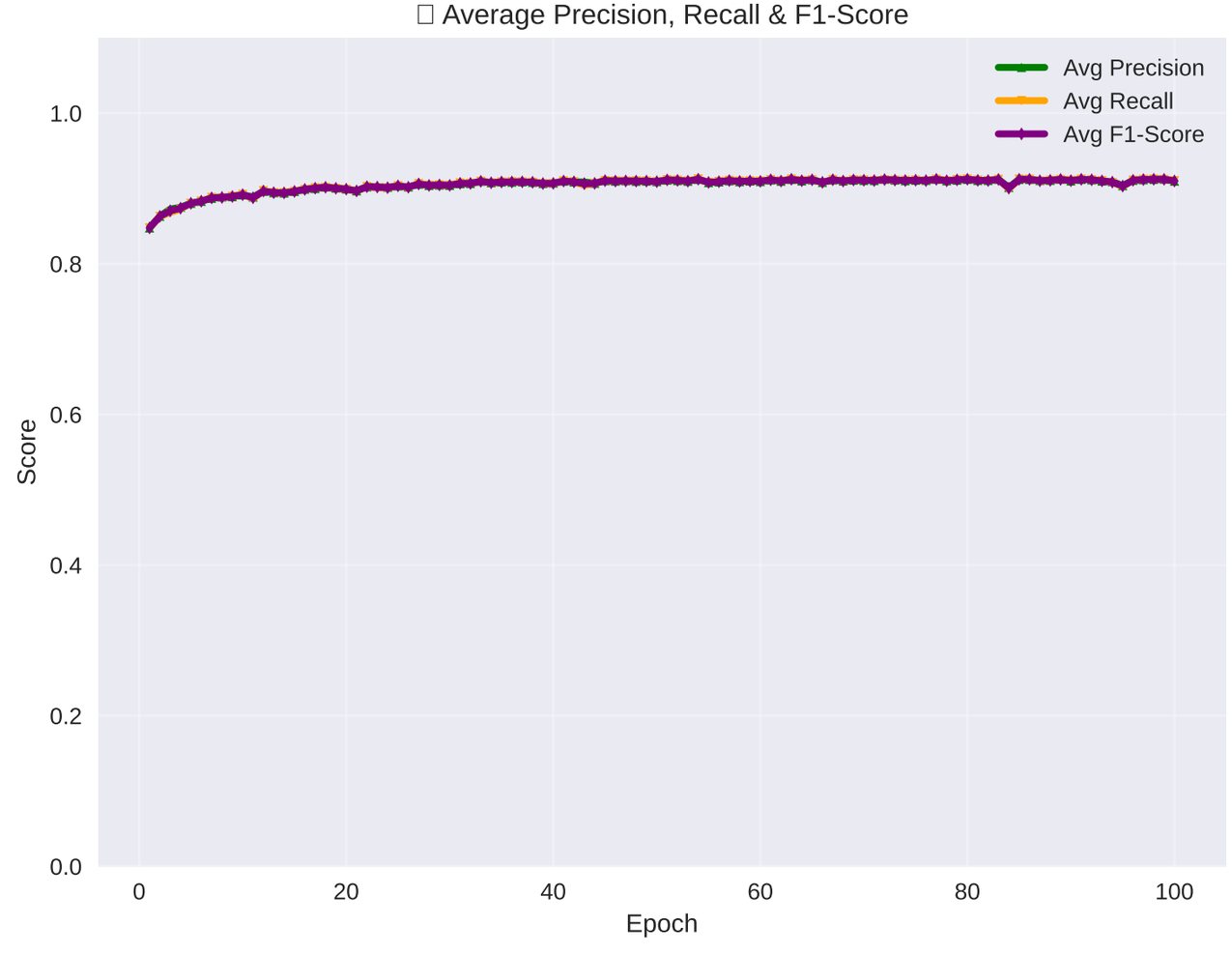
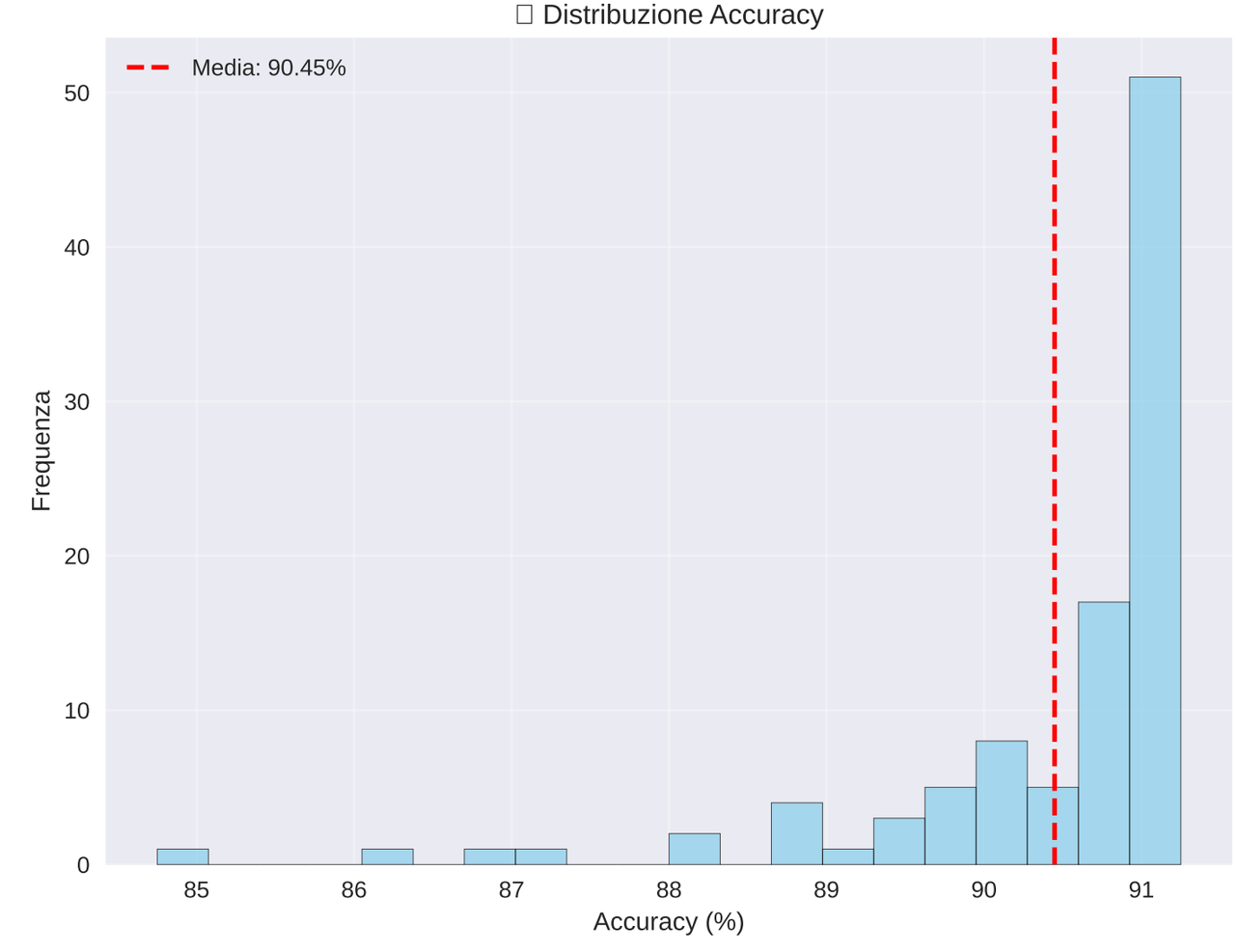
Federated Learning - Main Metrics Overview

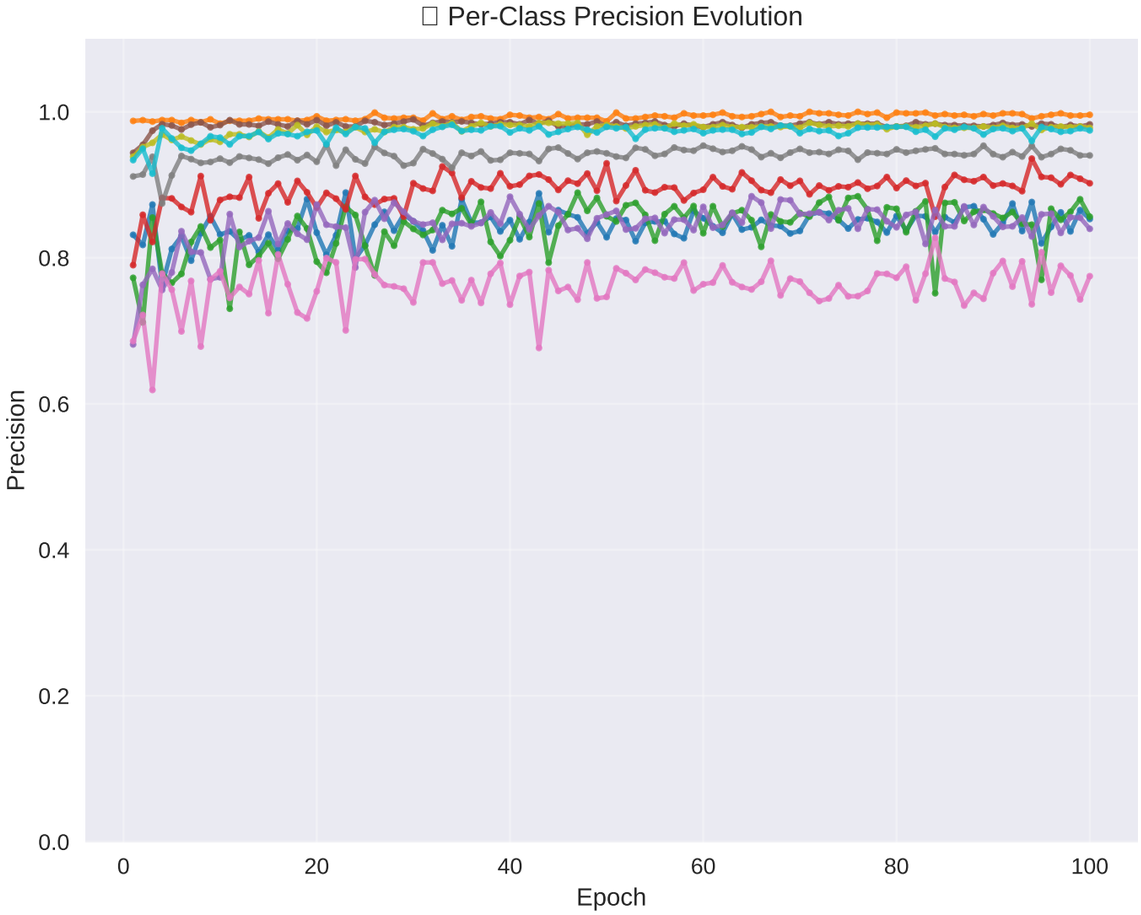


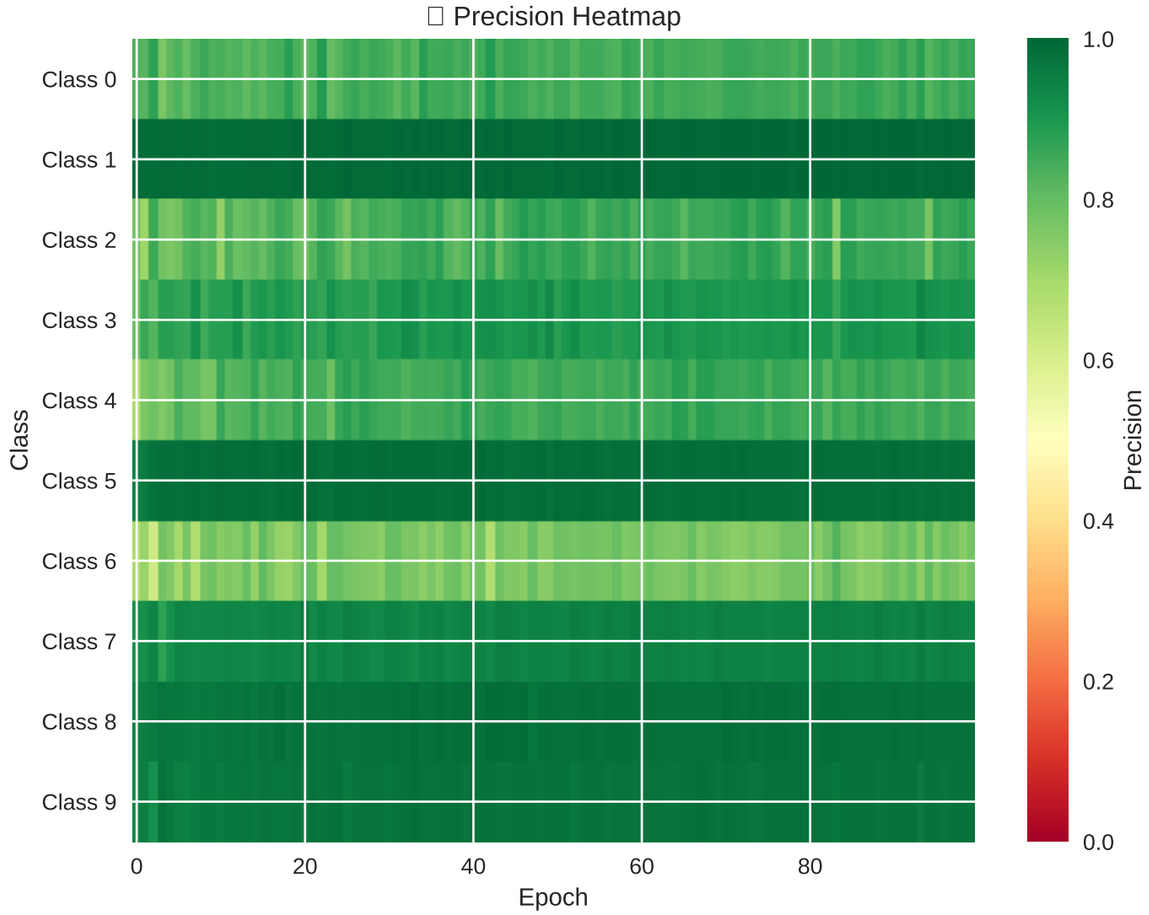


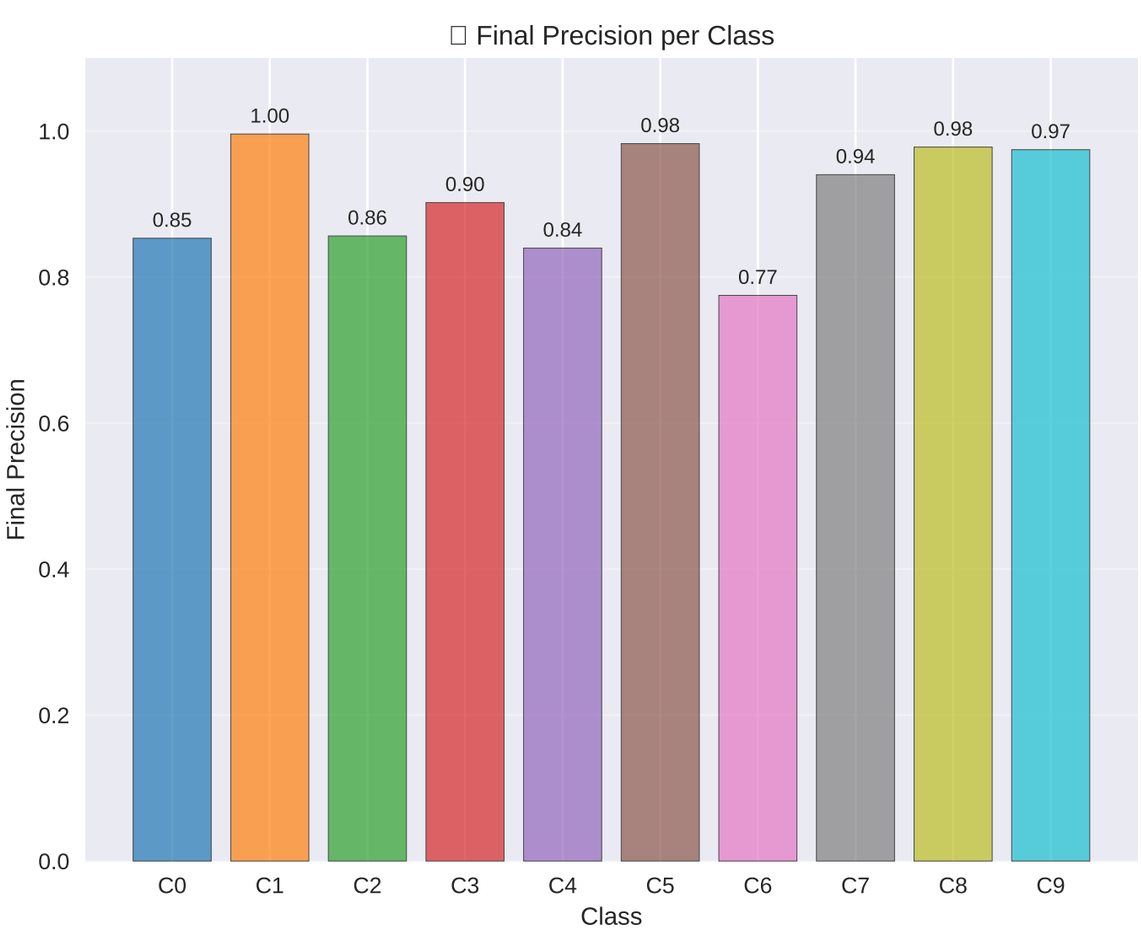


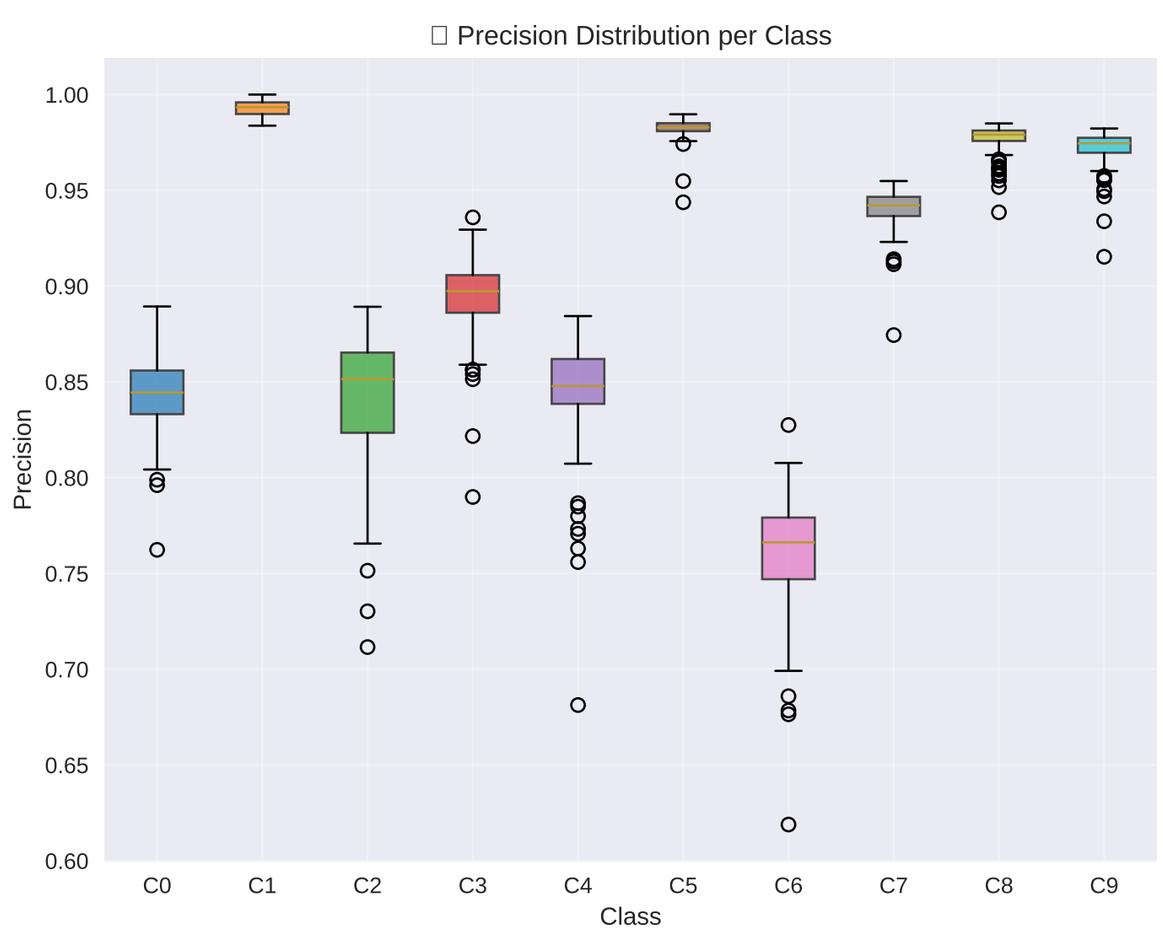


Per-Class Precision Analysis



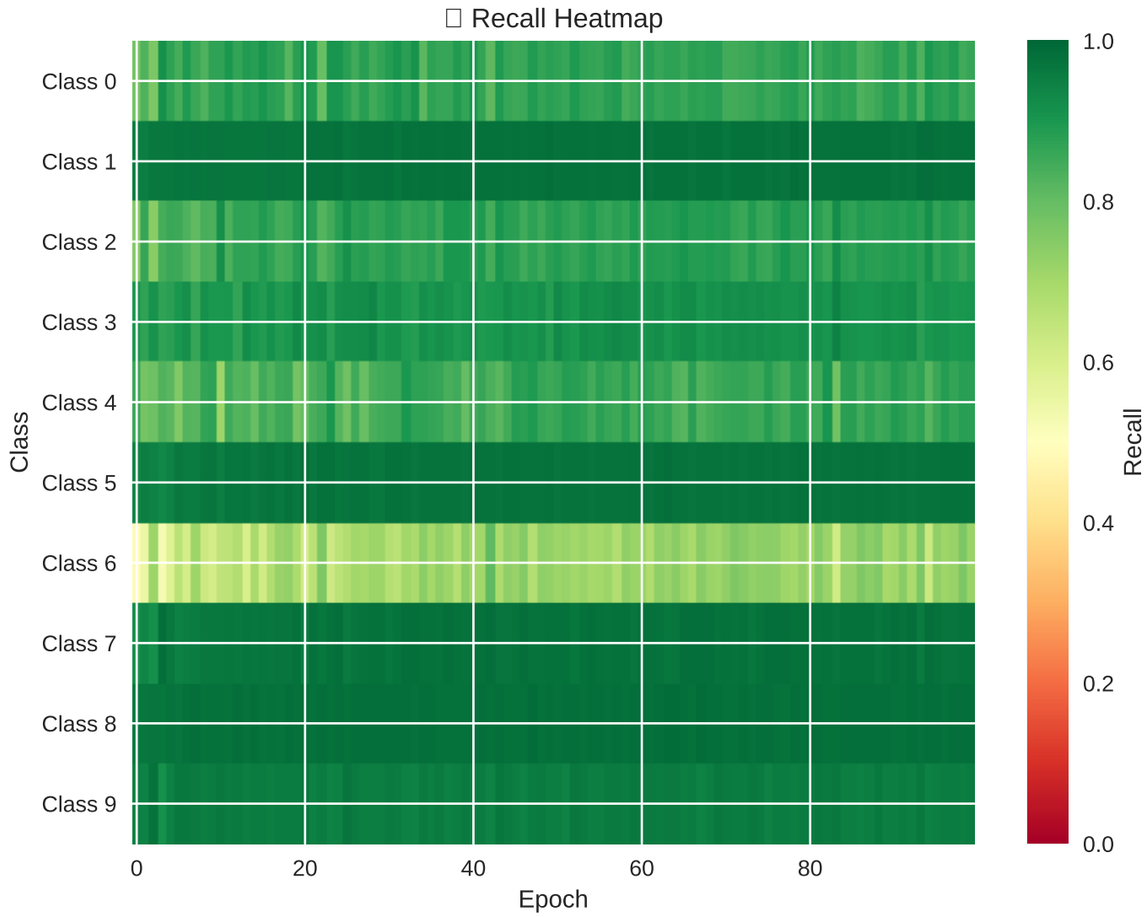


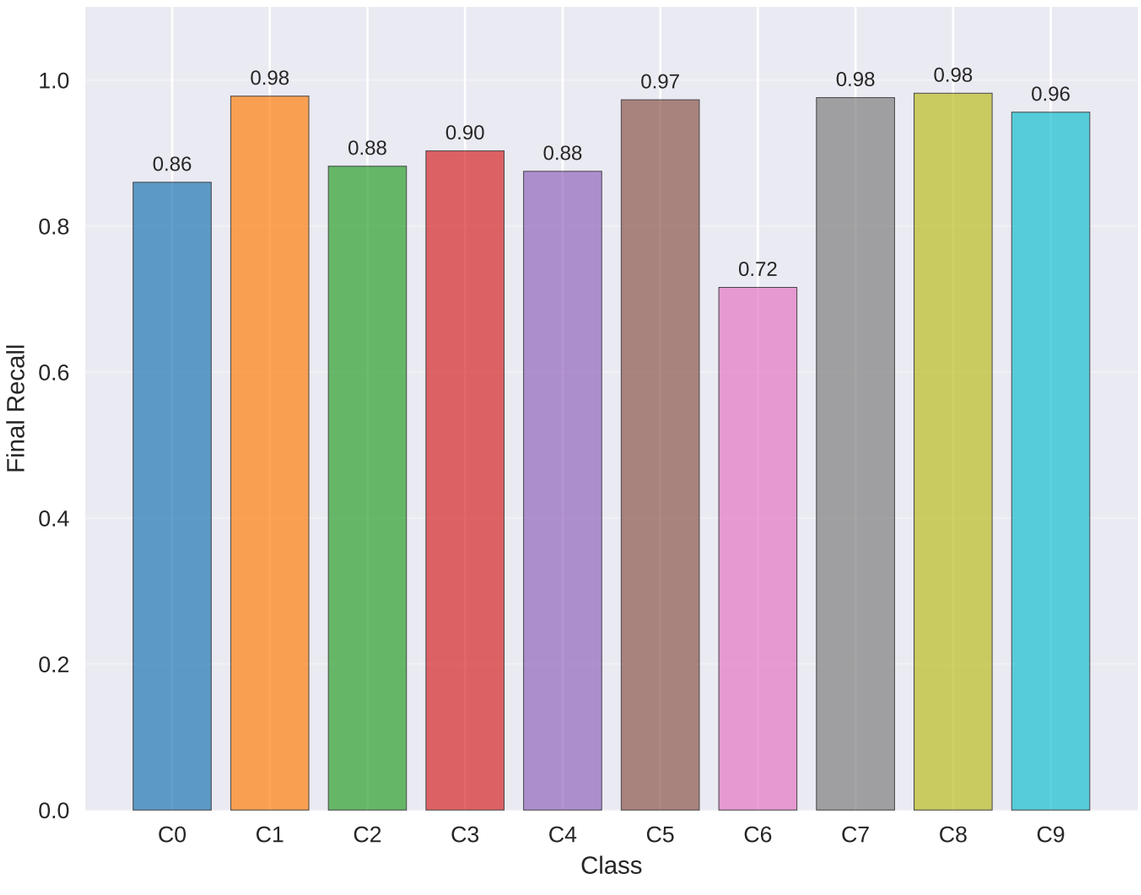


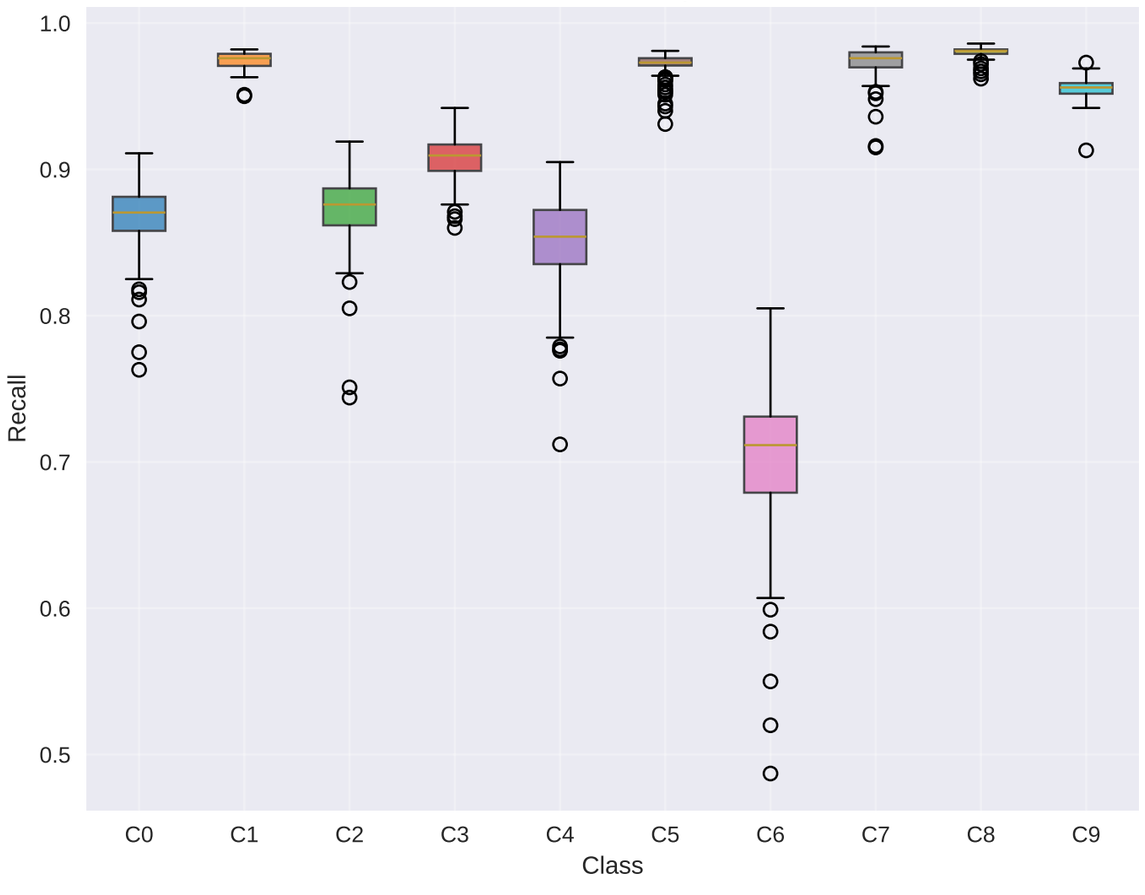


Per-Class Recall Analysis

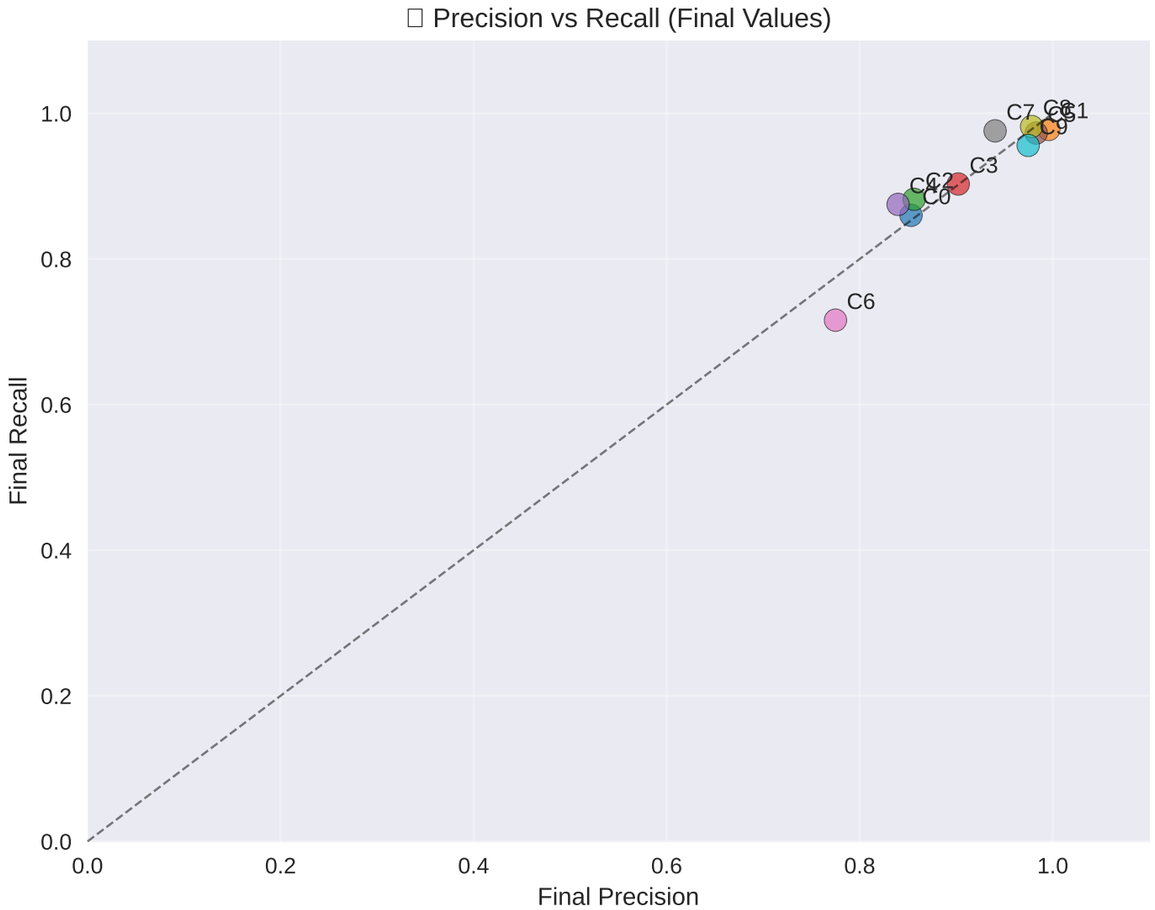






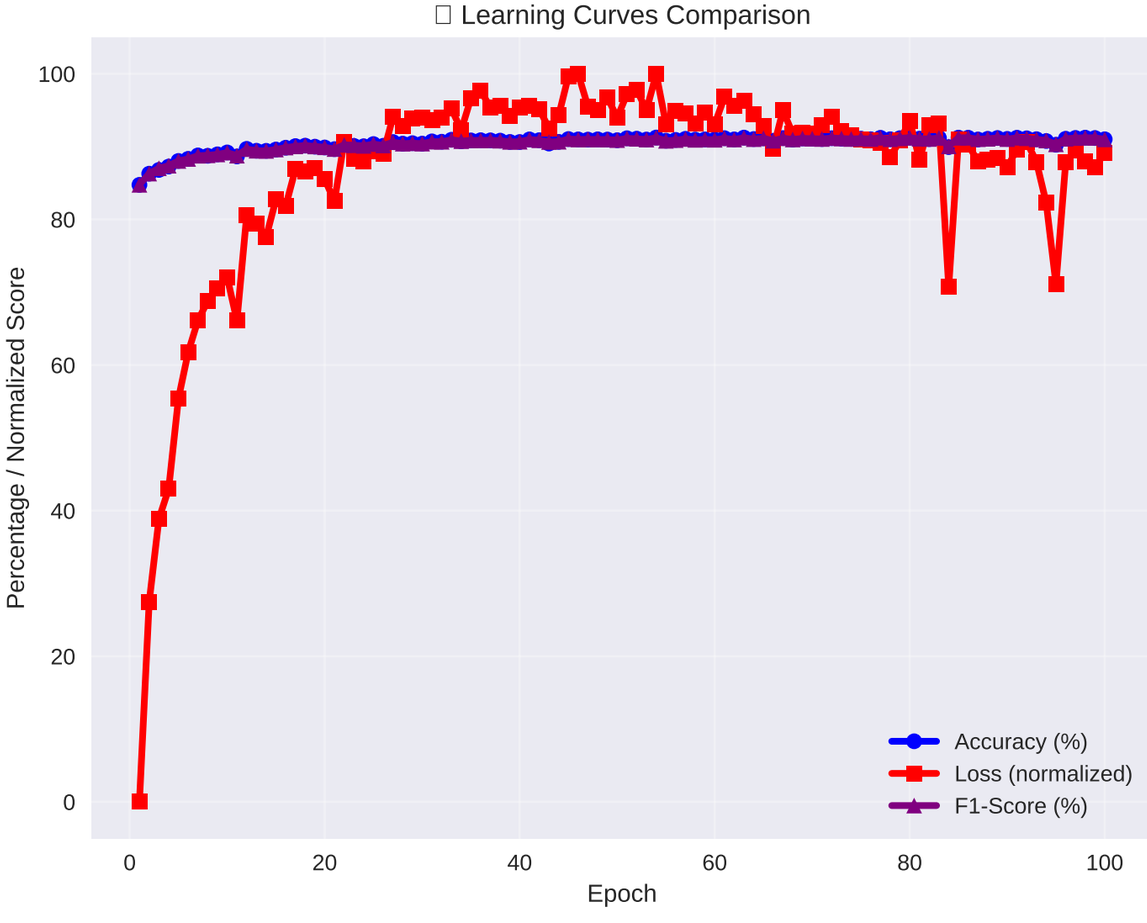


Comparative Analysis









Performance Summary Report

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FEDERATED LEARNING - PERFORMANCE SUMMARY
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FILE ANALIZZATO: 3000_results.csv
DATA ANALISI: 2025-08-31 21:38:01
EPOCHE TOTALI: 100

METRICHE FINALI:
• Accuracy: 91.01%
• Loss: 2.7391
• Average Precision: 0.910
• Average Recall: 0.910
• Average F1-Score: 0.910

EVOLUZIONE TRAINING:
• Miglioramento Accuracy: +6.26%
• Riduzione Loss: +1.5547
• Stabilità Accuracy (CV): 0.0125

PERFORMANCE PER CLASSE:
• Migliore Precision: Classe 1 (0.996)
• Peggiore Precision: Classe 6 (0.775)
• Migliore Recall: Classe 8 (0.982)
• Peggiore Recall: Classe 6 (0.716)

ANALISI DETTAGLIATA PER CLASSE:
• Classe 0: P=0.853, R=0.860, F1=0.857
• Classe 1: P=0.996, R=0.978, F1=0.987
• Classe 2: P=0.856, R=0.882, F1=0.869
• Classe 3: P=0.902, R=0.903, F1=0.903
• Classe 4: P=0.840, R=0.875, F1=0.857
• Classe 5: P=0.983, R=0.973, F1=0.978
• Classe 6: P=0.775, R=0.716, F1=0.744
• Classe 7: P=0.940, R=0.976, F1=0.958
• Classe 8: P=0.978, R=0.982, F1=0.980
• Classe 9: P=0.975, R=0.956, F1=0.965

VALUTAZIONE QUALITATIVA:
• Performance Globale: BUONA
• Stabilità Training: STABILE
• Balance Classes: SBILANCIATO

RACCOMANDAZIONI:
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